

1. Part 1: Exploring Sizing Tradeoffs Using SA

Design Specifications

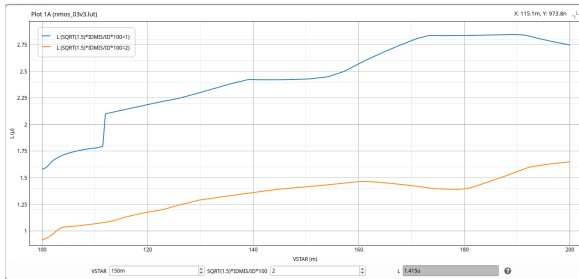
We want to design a simple current mirror with the following specs.

Parameter	Value
Current direction (source/sink)	Sink
Input Current	$10\mu\text{A}$
Output Current	$20\mu\text{A}$
% Change in Current for $\Delta V_{out} = 1\text{V}$	$< 10\%$
Percent mismatch: $\sigma(I_{out})/I_{out}$	$\leq 2\%$
Compliance voltage	$\leq 150\text{mV}$
Area	Minimize

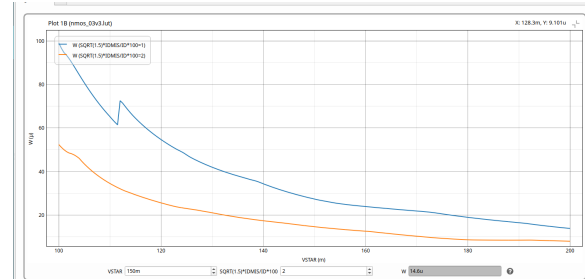
Device Type

Q: Sinking current means which device type? NMOS or PMOS?

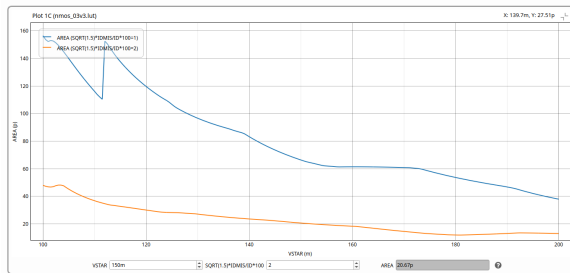
We require NMOS to sink current coming from analog block to ground.



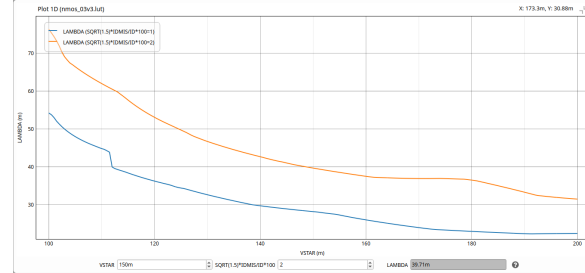
L vs V^*



W vs V^*



Area vs V^*



λ vs V^*

Figure 1: SA plots at constant swept mismatch $\sigma(I_{out})/I_{out}$ @1% and 2%

Channel Length Modulation

Q: The % Change in current translates to a spec on the $\lambda = 1/V_A$ of the device. How much is the required λ ?

$$\lambda = \Delta I_{out} / (I_{out} \Delta V_{out})$$

where V_{out} is V_{DS} of the output transistor, which translates λ to < 0.1

[23-07-2026 23:00:24] - Info: [Sizing Assistant] The resultant point 'sqrt(1.5)*idmis/id*100 = 5.652'

Mismatch at $\lambda = 0.1$

Name		TT-27.0
3	L	560n
4	W	5.64u

Initial Sizing at predefined λ

[23-07-2026 23:01:50] - Info: [Sizing Assistant] The resultant point 'LAMBDA = 39.41m'

Sizing with Mismatch = 2%

Name		TT-27.0
3	L	1.42u
4	W	14.6u

Sizing at predefined Mismatch

Figure 2: Device sizing under different design constraints

Comparison

From the figures above, We observe that sizing at Mismatch of 2% satisfies both requirements. We stick with Mismatch of 2% to minimize area.

Mismatch Comparison

Q: From the model file, find σ_{VT} and compare it to the value in ADT. Which one is higher, why?

The one from ADT is higher because it models all the possible mismatches in the transistor's parameters (u_n , C_{ox} , etc.) while the model file only models the mismatch in v_{th} (which is the dominant source of mismatch)

Design Trade-offs

Q: Does the results indicate that the highest V^* is desirable from the perspective of mismatch, area, and λ ? If yes, then the maximum V^* will be limited by the required compliance voltage.

highest V^* means low g_m/I_D and lower area (which is good) but suffers from more mismatch compared to when sized at high g_m/I_D . That is why we need to reduce W and increase L , providing high r_o for near ideal current copying and minimizing the area at the cost of higher V^* to keep the bias current the same.

2. Part 2: Current Mirror Simulation

2.1 1. Design and OP Analysis

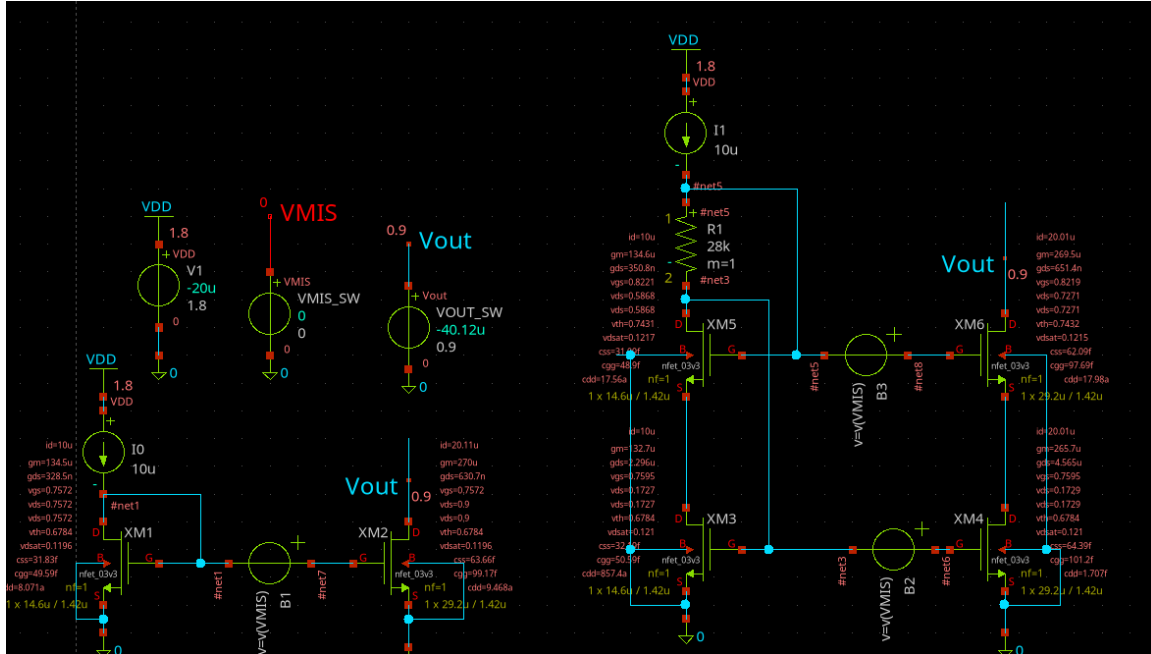
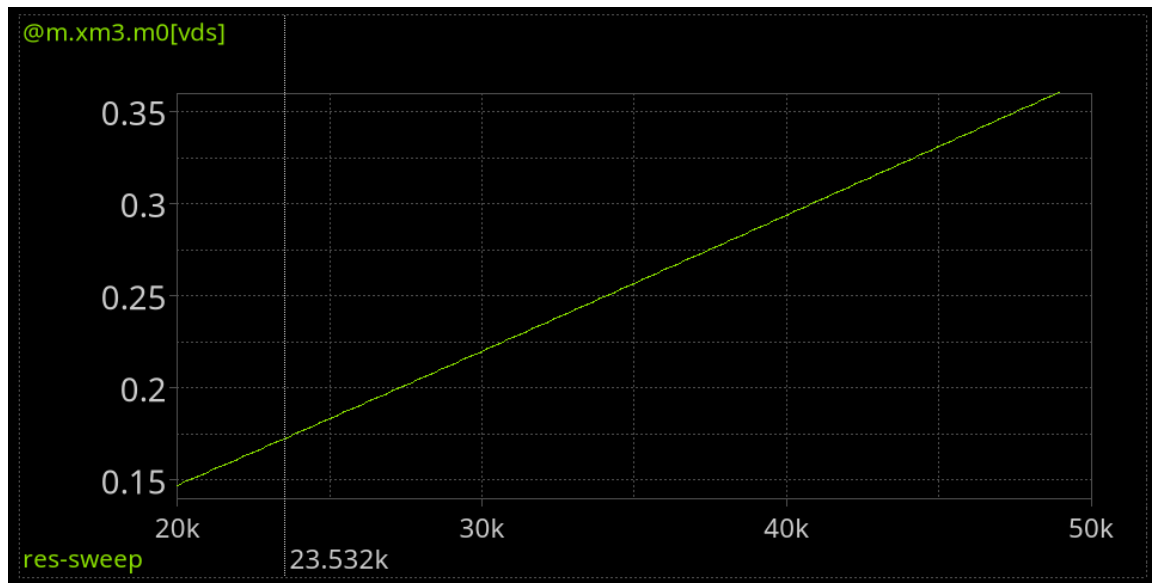


Figure 3: DC Operating points of Simple and Cascode Current Mirror

Analytical Calculation for R_B

Q: Assume we want to set a 50mV saturation margin for M2 and M3, i.e., $V_{DS2} \approx V_{DS3} \approx V^* + 50\text{mV}$. Ignore the body effect and calculate a rough value for R_B .

$$R_B = \frac{200\text{mV}}{10\mu\text{A}} = 20\text{k}\Omega$$

Figure 4: V_{DS3} vs R_B Sweep

R_B Selection

Q: Report V_{DS3} vs R_B . Choose R_B to satisfy the 50mV saturation margin requirement. Is the selected R_B value larger or smaller than the rough analytical value? Why?

I chose $R_B = 23.532k\Omega$, achieving $V_{DS3} = 172.7mV$

Saturation Check

Q: Do all transistors operate in saturation?

Yes, all transistors (XM1-6) are in saturation.

2.2 2. DC Sweep (I_{out} vs V_{OUT})

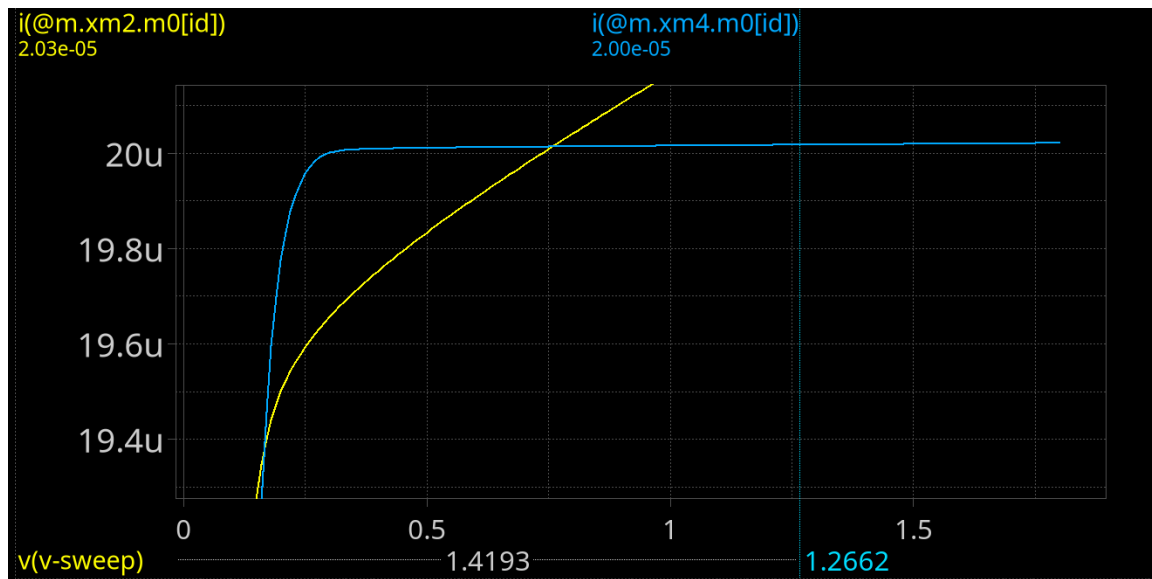


Figure 5: I_{out} vs V_{OUT} for the two CMs overlaid

Circuit Comparison

Comment on the difference between the two circuits.

as V_{out} changes in both circuits, We observe that current varies most in simple CM due to significantly lower R_{out} compared to that of cascode.

Compliance Voltage

Q: From the plot, find an estimate for the compliance voltage of each current mirror.

From the figures shown, Cascode CM requires higher compliance voltage to work as a current source compared to simple CM due to stacking transistors, where each one requires V^* to remain in saturation. the Cascode requires $246.1mV$ while simple requires $126.24mV$

Exact Output Current

Q: I_{out} of the simple CM is exactly equal to $I_B \times 2$ at a specific value of V_{OUT} . Why?

AT $V_{out}=0.7365V$. Because it is the exact value of V_{DS1} meaning same λ for both transistors.

Percent Change in Current

Q: Calculate the percent change in I_{out} when V_{OUT} changes from 0.5V to 1.5V (i.e., 1V change). Compare the simple mirror result to the value expected from Part 1.

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Using SPARSE 1.3 as Direct Linear Solver
Reference value : 1.51000e+00
No. of Data Rows : 181
Simple Current Mirror Simulation Results
i1          = 1.983370e-05
i2          = 2.046595e-05
perc_change = 3.137744e+00
Wide Swing Current Mirror Simulation Results
i1          = 2.001105e-05
i2          = 2.001923e-05
perc_change = 4.086906e-02

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Figure 6: Output current at $V_1=0.5V$ and $V_2=1.5V$

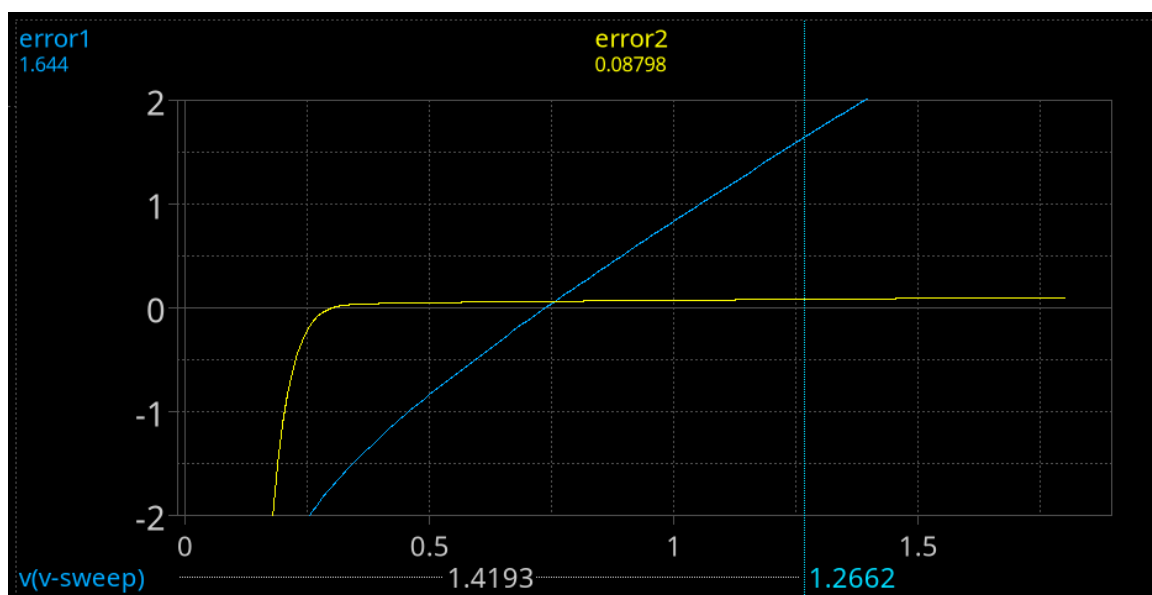
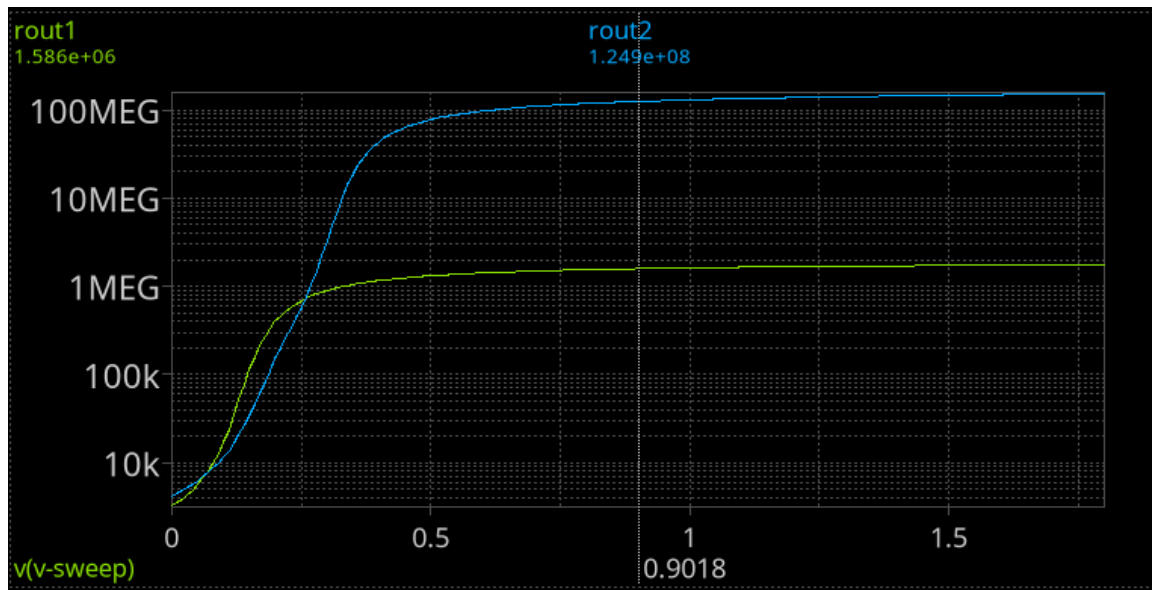


Figure 7: Percent of error in I_{out} vs V_{OUT} for the two CMs

Figure 8: R_{out} vs V_{OUT} for the two CMs

Output Resistance Extraction

Q: Report R_{out} vs V_{OUT} for the two CMs in the operating region overlaid in the same plot. Use log scale on the y-axis. Add a cursor at $V_{OUT} = V_{DD}/2$. Comment on the difference between the two circuits.

R_{out2} is greater than R_{out1} by order of $g_m r_o$. However, It comes at the cost of higher compliance voltage. As shown, the cascode needs higher voltage to flatten out compared to simple CM which gives constant R_{out} at much lower v_{out} .

R_{out} Variation

Q: Does R_{out} change with V_{OUT} ? Why?

Yes. At low V_{OUT} , the transistor operates in the triode region, resulting in a very low R_{out} . Once V_{OUT} increases enough to push the transistor into the saturation region, R_{out} jumps to a much higher value, but it continues to vary slightly due to channel-length modulation (the effective channel length shrinks as V_{DS} increases, subtly altering r_o).

Analytical vs Simulated R_{out}

Q: Analytically calculate R_{out} of both circuits at $V_{OUT} = V_{DD}/2$. Compare with simulation results in a table.

For a single-transistor current mirror, the output resistance is simply the intrinsic small-signal resistance (r_o).

$$g_{ds} = 630.5 \text{ nS}$$

$$R_{out1} = r_{o2} = \frac{1}{g_{ds}}$$

$$R_{out1} = \frac{1}{630.5 \times 10^{-9} \text{ S}} = 1.586 \text{ M}\Omega$$

Comment: This calculation perfectly matches the 1.586e+06 reading

2. Cascode Current Mirror (XM4 & XM6)

First, calculate the individual intrinsic resistances ($r_o = 1/g_{ds}$):

$$r_{o6} = \frac{1}{651 \times 10^{-9} \text{ S}} = 1.536 \text{ M}\Omega$$

$$r_{o4} = \frac{1}{4.565 \times 10^{-6} \text{ S}} = 219.06 \text{ k}\Omega$$

The exact analytical formula for cascode output resistance is:

$$R_{out2} = r_{o6} + r_{o4} + (g_{m6}) \cdot r_{o6} \cdot r_{o4}$$

Calculating the standard cascode gain

$$R_{out2} \approx 1.536 \text{ M}\Omega + 219 \text{ k}\Omega + (269.5 \times 10^{-6}) \cdot (1.536 \times 10^6) \cdot (219.06 \times 10^3)$$

$$R_{out2} \approx 1.75 \text{ M}\Omega + 90.68 \text{ M}\Omega = 92.43 \text{ M}\Omega$$

Comment: The analytical calculation without the body effect yields roughly 92.4 MΩ. However, the simulator reported 125 MΩ. Because the bulk of XM6 is tied to ground while its source voltage is 0.1729 V, it triggers the body effect (g_{mb6})

2.3 3. Mismatch

Mirror Device Mismatch (Simulated)

Q: Perform DC sweep for VMIS_SW from 0 to $\sqrt{1.5} \times v_{gmis}$ (one-sigma variation) and set VMIS_SW2 = 0. Find the percent change in I_{out} .

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No. of Data Rows : 18
Simple Current Mirror Simulation Results
i1          = 2.010552e-05 at= 0.000000e+00
i2          = 2.056807e-05 at= 1.700000e-03
perc_change = 2.274449e+00
Wide Swing Current Mirror Simulation Results
i1          = 2.001485e-05 at= 0.000000e+00
i2          = 2.046390e-05 at= 1.700000e-03
perc_change = 2.218695e+00
binary raw file "cmirror_dc.raw"
Doing analysis at TEMP = 27.000000 and TNOM = 27.000000

Using SPARSE 1.3 as Direct Linear Solver
Note: vmis_sw2: has no value, DC 0 assumed

No. of Data Rows : 18
"Wide Swing Current Mirror Simulation Results (Cascode
mismatch)": no such command available in ngspice
i1          = 2.001485e-05 at= 0.000000e+00
i2          = 2.002039e-05 at= 1.700000e-03
perc_change = 2.767562e-02
binary raw file "cmirror_dc.raw"

```

Figure 9

Mirror Device Mismatch (Analytical)

Q: Analytically calculate the percent change in I_{out} and compare it to the simulation result.

From ADT, it is found that v_{gms} at $W=14.6\mu$ and $L=1.42\mu$ to be $= 1.247m$. Using the formula, We get $1.7m$, accounting for variances in both transistors (M2 has double the area of M1)

$$\frac{\Delta I_{out}}{I_{out}} = \frac{g_{m2} \cdot \Delta V_{th}}{I_{out}}$$

$$\% \text{ Change} = \frac{270 \mu S \cdot 1.7 \text{ mV}}{20.01 \mu A} \times 100 = 2.2935\%$$

As for the cascode:

$$\frac{\Delta I_{out}}{I_{out}} = \frac{g_{m4} \cdot \Delta V_{th}}{I_{out}}$$

$$\% \text{ Change} = \frac{265.7 \mu S \cdot 1.7 \text{ mV}}{20.01 \mu A} \times 100 = 2.257\%$$

Note on Mismatch Dominance:

The mismatch voltage sweep (V_{MIS}) applied in the simulation primarily models the threshold voltage variation (ΔV_{th}) between the matched transistor pair

Cascode Device Mismatch (Analytical)

Q: Analytically calculate the percent change in I_{out} and compare it to the simulation result.

$$\frac{\Delta I_{out}}{I_{out}} \approx \frac{g_{ds4} \cdot \Delta V_{th}}{I_{out}}$$

$$\% \text{ Change} \approx \frac{4.565 \mu\text{S} \cdot 1.7 \text{ mV}}{20.01 \mu\text{A}} \times 100 = 0.0388\%$$

I assumed here that $g_m r_o \gg 1$, leading to the simplification to $1/r_o$

Mismatch Contribution Comparison

Q: Which mismatch contribution is more pronounced? Why?

The mirroring transistors are the primary contributors (M1,M2,M3,M4) as the small signal perturbation in the gate voltage is multiplied by g_m in the output current.

Design Trade-offs for Mismatch

Q: For a given total area: Which design decision is better: setting the same W and L for the mirror and cascode devices? Or using larger W and L for the current mirror devices? Why?

To minimise area, We could decrease both W and L of M5 and M6 by the same ratio. since the small mismatch error at the gate is severely downgraded due to the degeneration resistance of M4 (i.e this error will change to small signal current $\frac{g_m}{1+g_m r_o} * v_{mis}$). Furthermore, We can increase both W and L of M1-4 to decrease mismatch, but that comes at the cost of area.

2.4 4. Monte Carlo (MC) Simulation

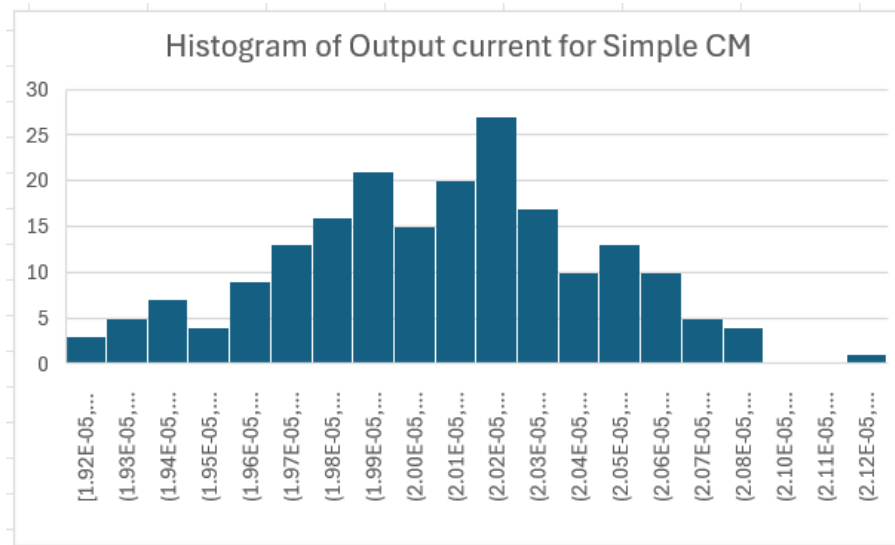


Figure 10: Histogram of I_{out} from Monte Carlo simulation for simple CM

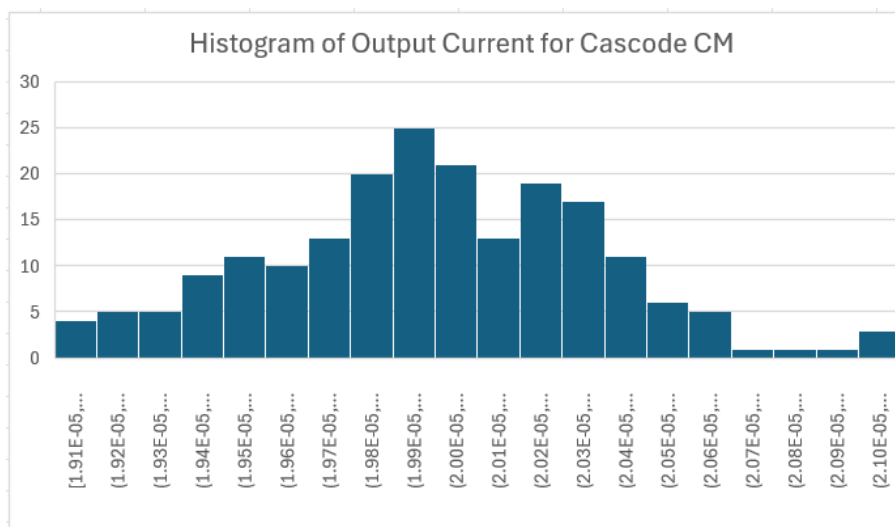


Figure 11: Histogram of I_{out} from Monte Carlo simulation for cascode CM

Monte Carlo Results

Q: Calculate the standard deviation percentage $\sigma(I_{out})/I_{out,avg}$.

Percent Mismatch for Simple Current Mirror is 1.98%,
where that of Cascode Current Mirror is 1.93%

Q: Compare the MC simulation result to the expected analytical result.

The results match my design requirements of less than 2% mismatch.